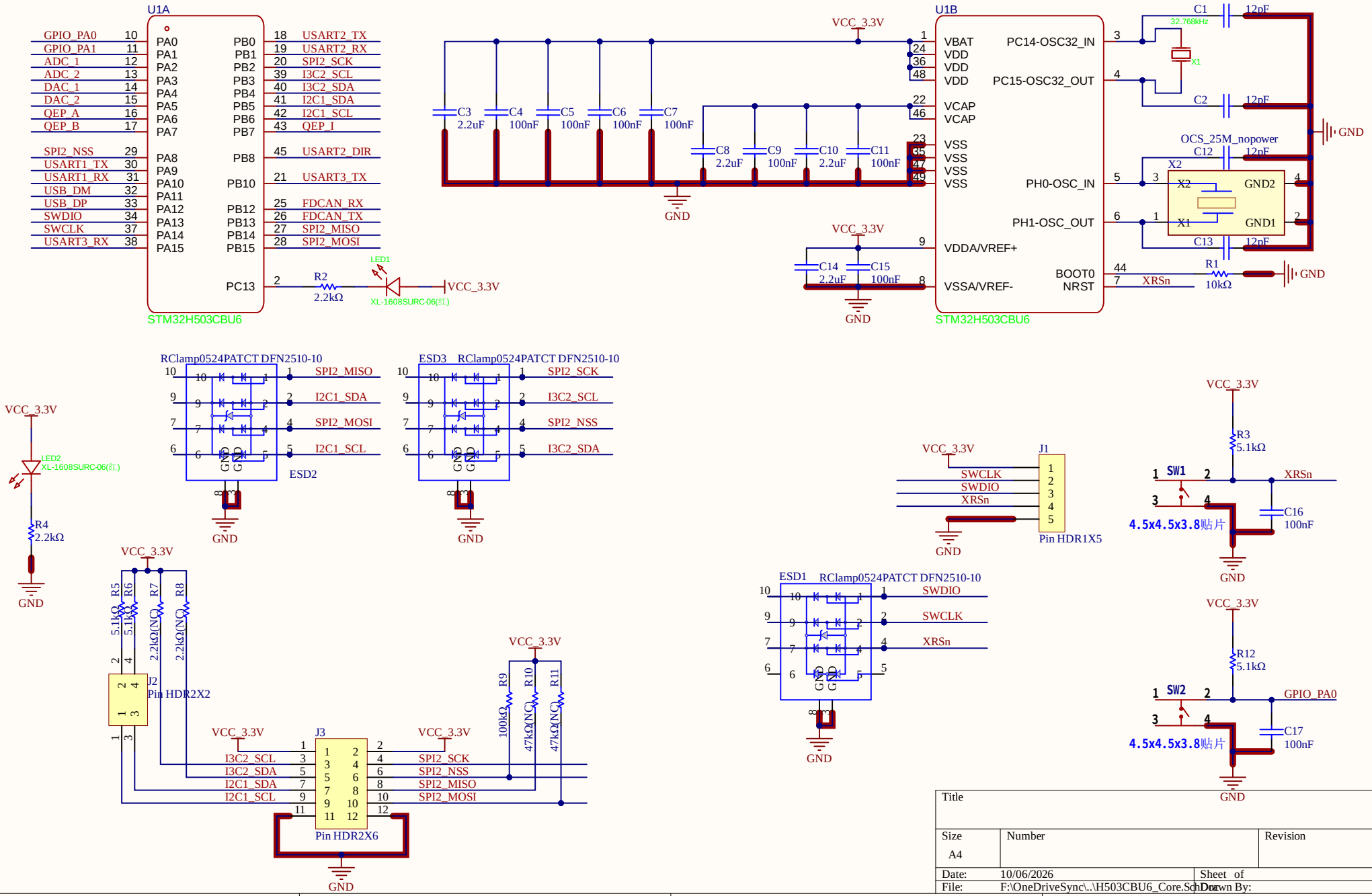
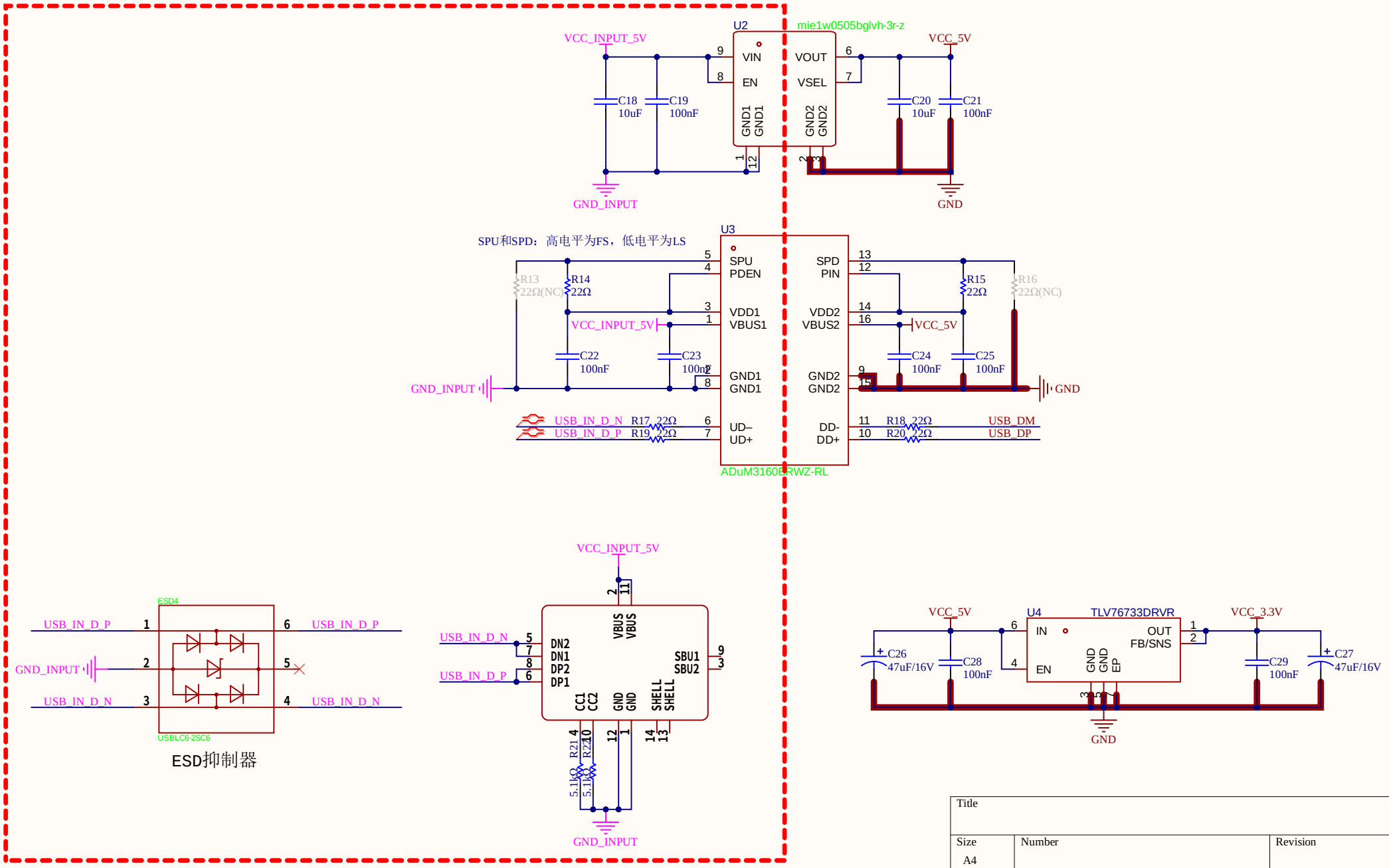
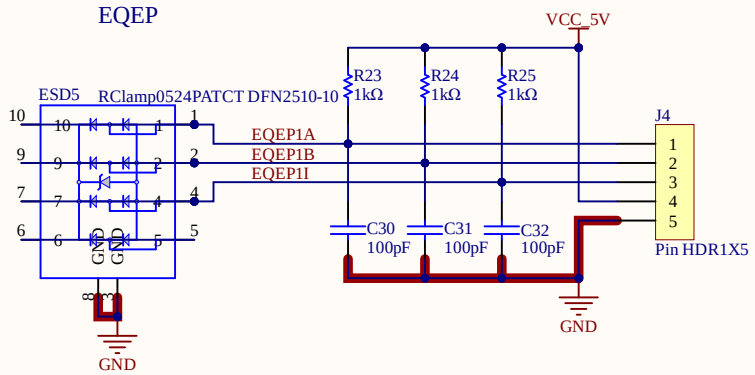
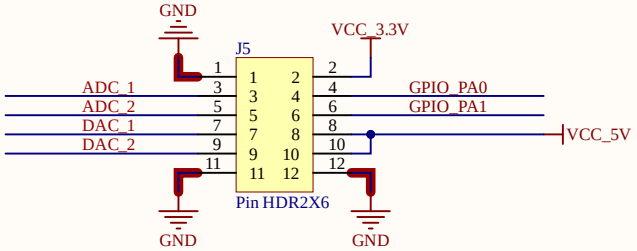
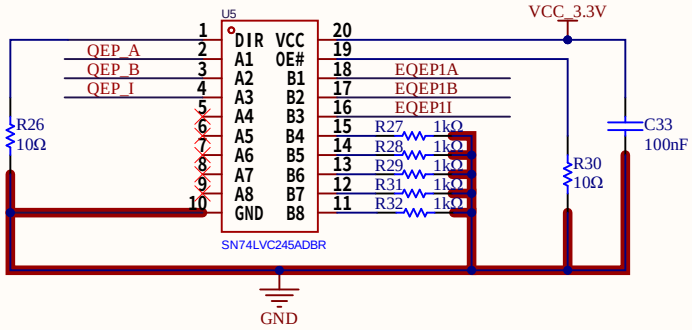

U1A
GPIO_PA0 10
GPIO_PA1 11
ADC_1 12
ADC_2 13
DAC_1 14
DAC_2 15
QEP_A 16
QEP_B 17
SPI2_NSS 29
USART1_TX 30
USART1_RX 31
USB_DM 32
USB_DP 33
SWDIO 34
SWCLK 37
USART3_RX 38
PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7 PA8 PA9 PA10 PA11 PA12 PA13 PA14 PA15
PB0 PB1 PB2 PB3 PB4 PB5 PB6 PB7 PB8 PB10 PB12 PB13 PB14 PB15
PC13
18 USART2_TX
19 USART2_RX
20 SPI2_SCK
39 I3C2_SCL
40 I3C2_SDA
41 I2C1_SDA
42 I2C1_SCL
43 QEP_I
45 USART2_DIR
21 USART3_TX
25 FDCAN_RX
26 FDCAN_TX
27 SPI2_MISO
28 SPI2_MOSI
STM32H503CBU6
R2
2.2kΩ
LED1
XL-1608SURC-06(红)
VCC_3.3V
U1B
VBAT
VDD
VDD
VDD
VCAP
VCAP
VSS
VSS
VSS
VSS
VDDA/VREF+
VSSA/VREF-
PC14-OSC32_IN
PC15-OSC32_OUT
PH0-OSC_IN
PH1-OSC_OUT
BOOT0
NRST
XRSn
C1 12pF
32.768kHz
X1
C2 12pF
OCS_25M_nopower
C12 12pF
X2
GND2
GND1
C13 12pF
R1
10kΩ
C3 2.2uF
C4 100nF
C5 100nF
C6 100nF
C7 100nF
C8 2.2uF
C9 100nF
C10 2.2uF
C11 100nF
C14 2.2uF
C15 100nF
GND
STM32H503CBU6
RClamp0524PATCT DFN2510-10
ESD2
SPI2_MISO
I2C1_SDA
SPI2_MOSI
I2C1_SCL
ESD3
SPI2_SCK
I3C2_SCL
SPI2_NSS
I3C2_SDA
LED2
XL-1608SURC-06(红)
R4
2.2kΩ
J1
SWCLK
SWDIO
XRSn
Pin HDR1X5
R3
5.1kΩ
SW1
XRSn
C16
100nF
4.5x4.5x3.8贴片
ESD1
SWDIO
SWCLK
XRSn
R5 5.1kΩ
R6 5.1kΩ
R7 2.2kΩ(NC)
R8 2.2kΩ(NC)
J2
Pin HDR2X2
J3
I3C2_SCL
I3C2_SDA
I2C1_SDA
I2C1_SCL
SPI2_SCK
SPI2_NSS
SPI2_MISO
SPI2_MOSI
Pin HDR2X6
R9 100kΩ
R10 47kΩ(NC)
R11 47kΩ(NC)
R12
5.1kΩ
SW2
GPIO_PA0
C17
100nF
4.5x4.5x3.8贴片
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Size
A4
Number
Revision
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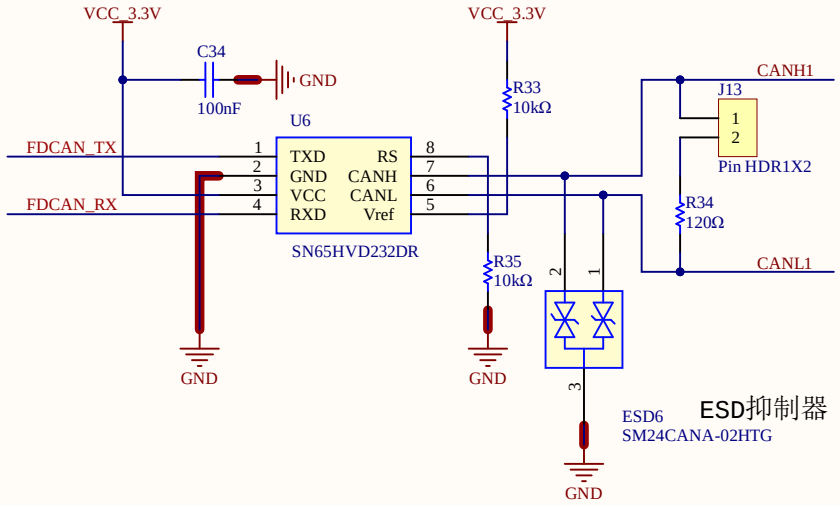
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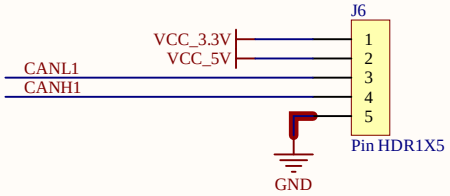




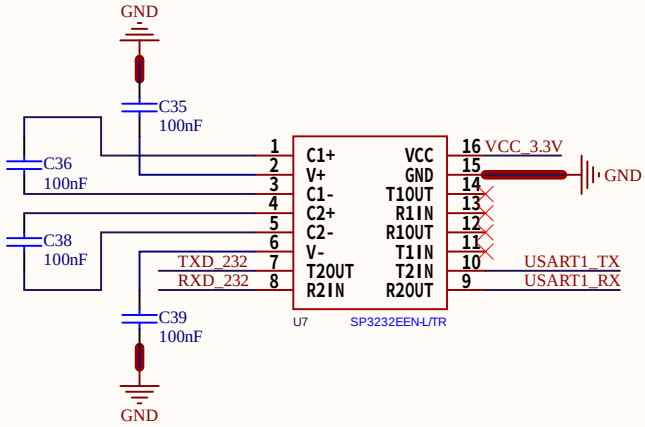


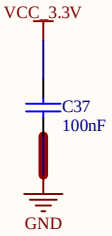
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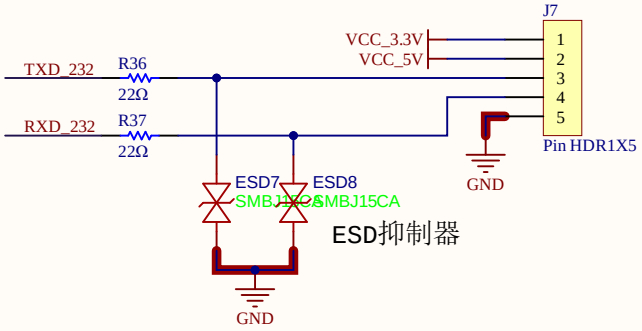


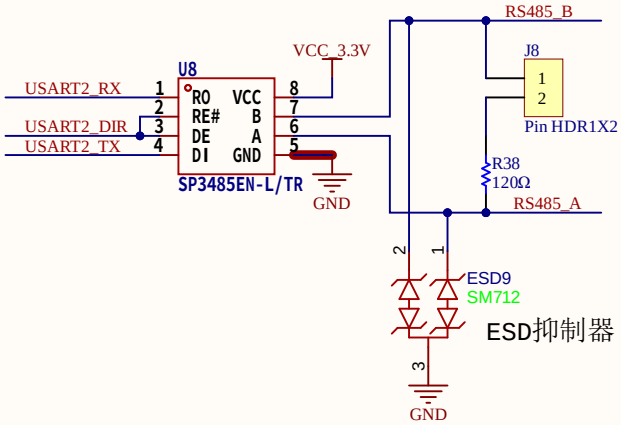


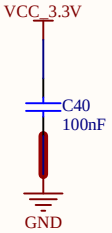
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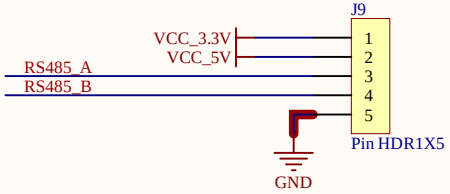


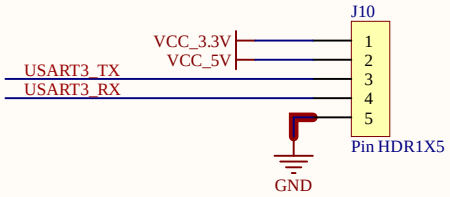




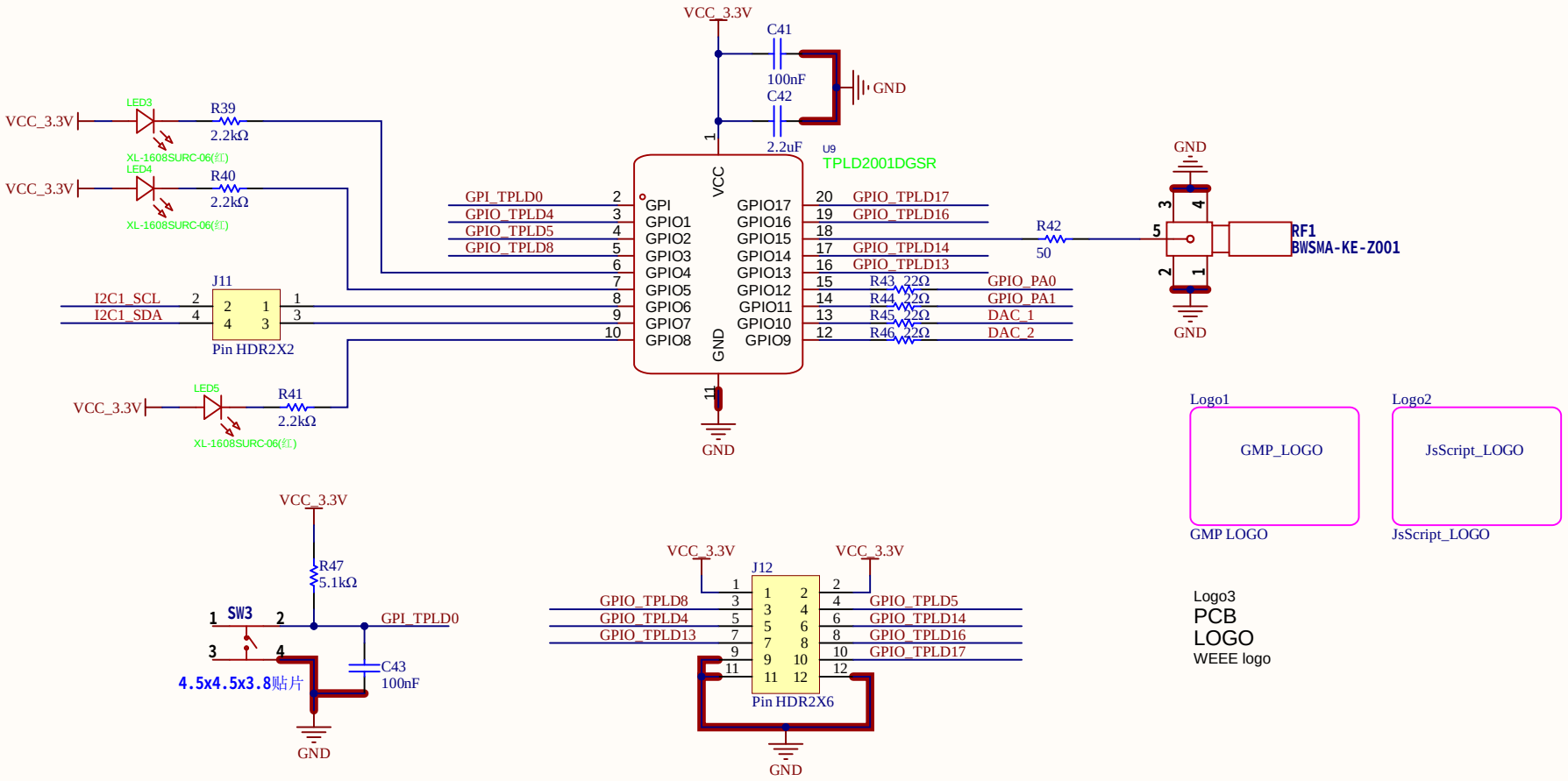




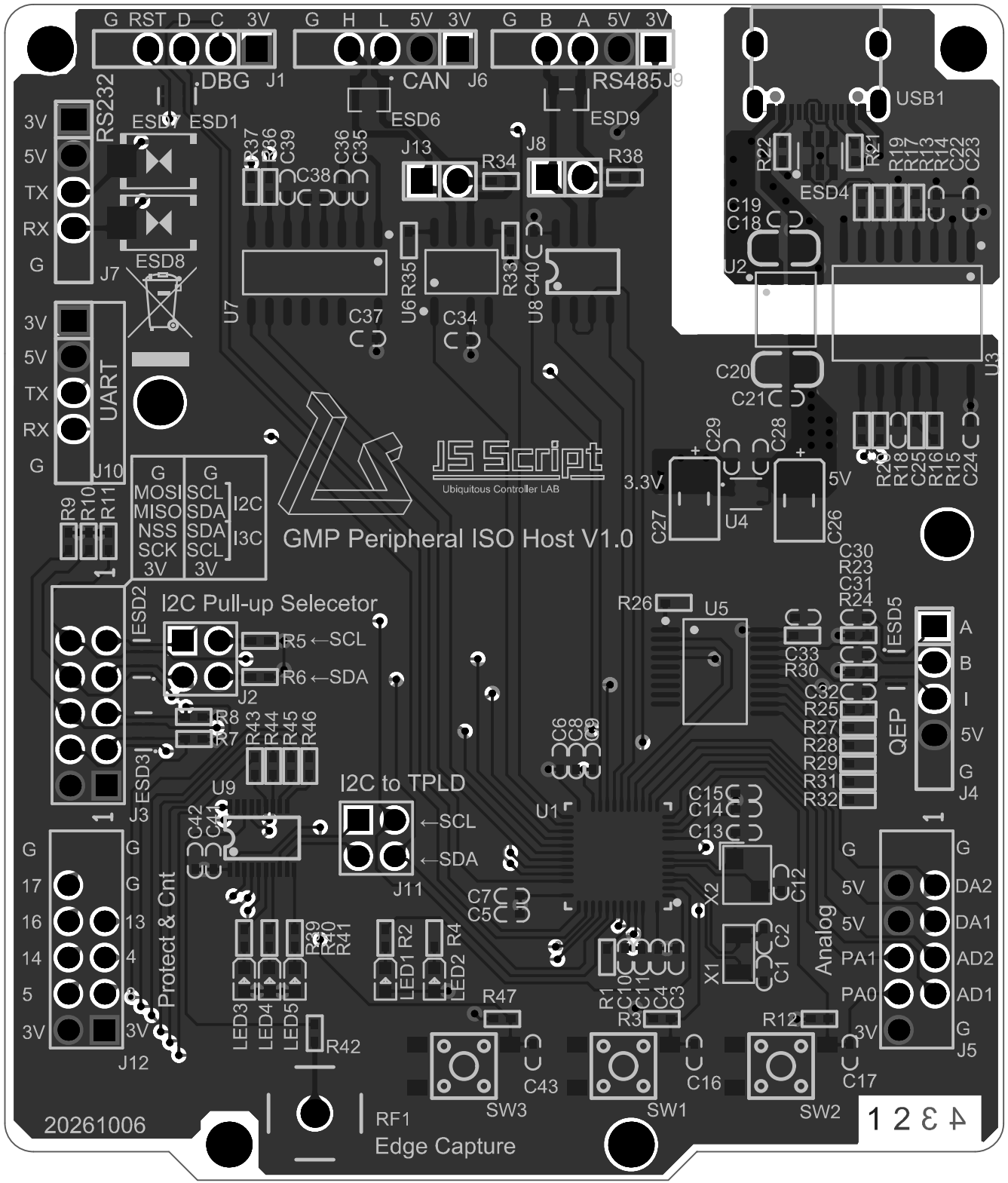




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Size: A4	Number	Revision
Date: 10/06/2026		Sheet of
File: F:\OneDriveSync\..\TPLD.SchDoc		Drawn By:


GMP Peripheral ISO Host V1.0
Ubiquitous Controller LAB
20261006
Edge Capture
I2C Pull-up Selecetor
I2C to TPLD
Protect & Cnt
Analog
QEP
RS232
UART
DBG J1
CAN J6
RS485 J9
USB1
ESD4
3.3V
5V
SW1
SW2
SW3
RF1

Line #	Name	Description	Designator	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	12pF	无极性电容	C1, C2, C12, C13	4							
	2.2uF	无极性电容	C3, C8, C10, C14, C42	5							
	100nF	无极性电容	C4, C5, C6, C7, C9, C11, C15, C16, C17, C19, C21, C22, C23, C24, C25, C28, C29, C33, C34, C35, C36, C37, C38, C39, C40, C41, C43	27							
	10uF	无极性电容	C18, C20	2							
	47uF/16V	C136751	C26, C27	2							
	100pF	无极性电容	C30, C31, C32	3							
	RClamp0524PATCT DFN2510-10		ESD1, ESD2, ESD3, ESD5	4							
	USBLC6-2SC6	ESD 4P USB	ESD4	1							
	SM24CANA-02HTG		ESD6	1							
	SMBJ15CA	极性:双向;反向截止电压(Vrwm):15V;钳位电压:24.4V;峰值脉冲电流(Ipp):24.59A@10/1000us;峰值脉冲功率(Ppp):600W@10/1000us;击穿电压(VBR):18.5V;C190777570	ESD7, ESD8	2							
	SM712	极性:双向;反向截止电压(Vrwm):12V，7V;钳位电压:23V;26V;峰值脉冲电流(Ipp):12A;峰值脉冲功率(Ppp):300W;击穿电压(VBR):7.5V，13.3V;RS485专用总线保护器。;C7420375	ESD9	1							
	Pin HDR1X5		J1, J4, J6, J7, J9, J10	6							
	Pin HDR2X2		J2, J11	2							
	Pin HDR2X6		J3, J5, J12	3							
	Pin HDR1X2		J8, J13	2							
	XL-1608SURC-06(红)	LED 0603	LED1, LED2, LED3, LED4, LED5	5							
	GMP LOGO	LOGO GMP Main.	Logo1	1							
	JsScript_LOGO	LOGO JsScript	Logo2	1							
	10kΩ		R1, R33, R35	3							
	2.2kΩ		R2, R4, R39, R40, R41	5							
	5.1kΩ		R3, R5, R6, R12, R21, R22, R47	7							
	2.2kΩ(NC)		R7, R8	2							
	100kΩ		R9	1							
	47kΩ(NC)		R10, R11	2							
	22Ω(NC)		R13, R16	2							
	22Ω		R14, R15, R17, R18, R19, R20, R36, R37, R43, R44, R45, R46	12							
	1kΩ		R23, R24, R25, R27, R28, R29, R31, R32	8							
	10Ω		R26, R30	2							
	120Ω		R34, R38	2							
	50		R42	1							
	BWSMA-KE-Z001	SMA	RF1	1							
	4.5x4.5x3.8贴片		SW1, SW2, SW3	3							
	STM32H503CBU6	CPU内核:ARM Cortex-M33;CPU最大主频:250MHz;CPU位数:32 Bit;C25503101	U1	1							
	mie1w0505bgl-3r-z	MIE1W0505BGL-VH是一款隔离式，稳压型DC-DC模块。它可支持3V至5.5V输入电压应用。MIE1W0505BGL-VH具有出色的负载调整率和线性调整率，支持高达1W的输出功率。 MIE1W0505BGL-VH将功率MOS、变压器和反馈电路集成在单芯片中，实现了出色的性能并节省了尺寸。 MIE1W0505BGL-VH支持稳压输出，当输出电压（Vout）下降且低于目标电压时，IC开始切换，将功率从输入电压（Vin）传输到输出电压（Vout），直到	U2	1							
	ADuM3160BRWZ-RL	全速/低速USB数字隔离器	U3	1							
	TLV76733DRVR	输出类型:固定;最大输入电压:16V;输出电压:3.3V;输出电流:1A;电源纹波抑制比(PSRR):70dB@(120Hz);输出极性:正极;	U4	1							
	SN74LVC245ADBR		U5	1							
	SN65HVD232DR		U6	1							
	SP3232EEN-L/TR		U7	1							
	SP3485EN-L/TR		U8	1							
	TPLD2001DGSR	TPLD2001DGSR	U9	1							
	TYPE-C 16PIN 2MD(073)	USB Type C 12P	USB1	1							
	32.768kHz	OSC 3215, 32.768kHz	X1	1							
	OCS_25M_nopower	OSC passive, 3225	X2	1							